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Chapter #6: Bipolar Junction Transistors

from **Microelectronic Circuits** Text

by Sedra and Smith

Oxford Publishing

Introduction

- **IN THIS CHAPTER YOU WILL LEARN**
  - The physical structure of the bipolar transistor and how it works.
  - How the voltage between two terminals of the transistor controls the current that flows through the third terminal, and the equations that describe these current-voltage relationships.
  - How to analyze and design circuits that contain bipolar transistors, resistors, and dc sources.
  - How the transistor can be used to make an amplifier.

Introduction

- **IN THIS CHAPTER YOU WILL LEARN**
  - How to obtain linear amplification from the fundamentally nonlinear BJT.
  - The three basic ways for connecting a BJT to be able to construct amplifiers with different properties.
  - Practical circuits for bipolar-transistor amplifiers that can be constructed by using discrete components.

Introduction

- This chapter examines another three-terminal device.
  - **bipolar junction transistor**
  - Presentation of this material mirrors chapter 5.
- BJT was invented in 1948 at Bell Telephone Laboratories.
  - Ushered in a new era of solid-state circuits.
  - It was replaced by MOSFET as predominant transistor used in modern electronics.

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6.1. **Device Structure and Physical Operation**

- Figure 6.1. shows simplified structure of BJT.
- Consists of three semiconductor regions:
  - **emitter** region (*n*-type)
  - **base** region (*p*-type)
  - **collector** region (*n*-type)
- Type described above is referred to as *npn*.
  - However, *pnp* types do exist.

6.1.1. Simplified Structure and Modes of Operation

- Transistor consists of two *pn*-junctions:
  - **emitter-base** junction (EBJ)
  - **collector-base** junction (CBJ)
- Operating mode depends on biasing.
  - **active** mode – used for amplification
  - **cutoff** and **saturation** modes – used for switching.

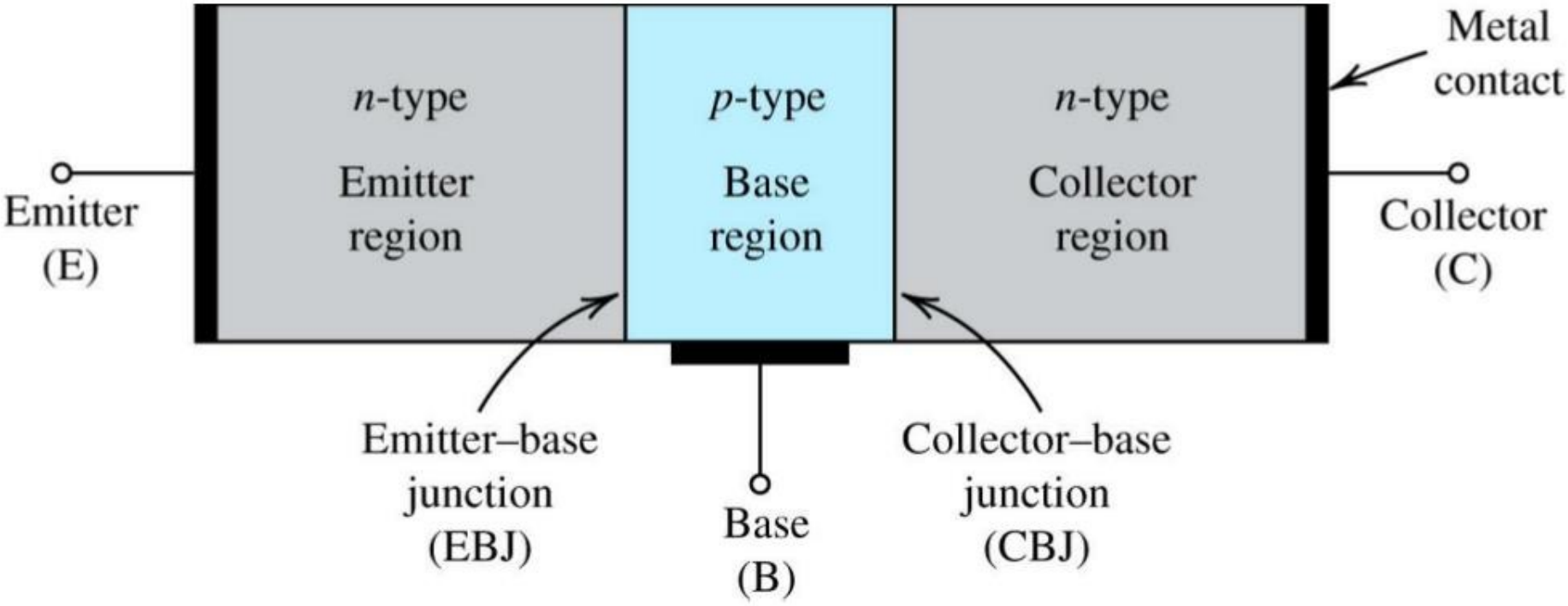


Figure 6.1: A simplified structure of the *npn* transistor.

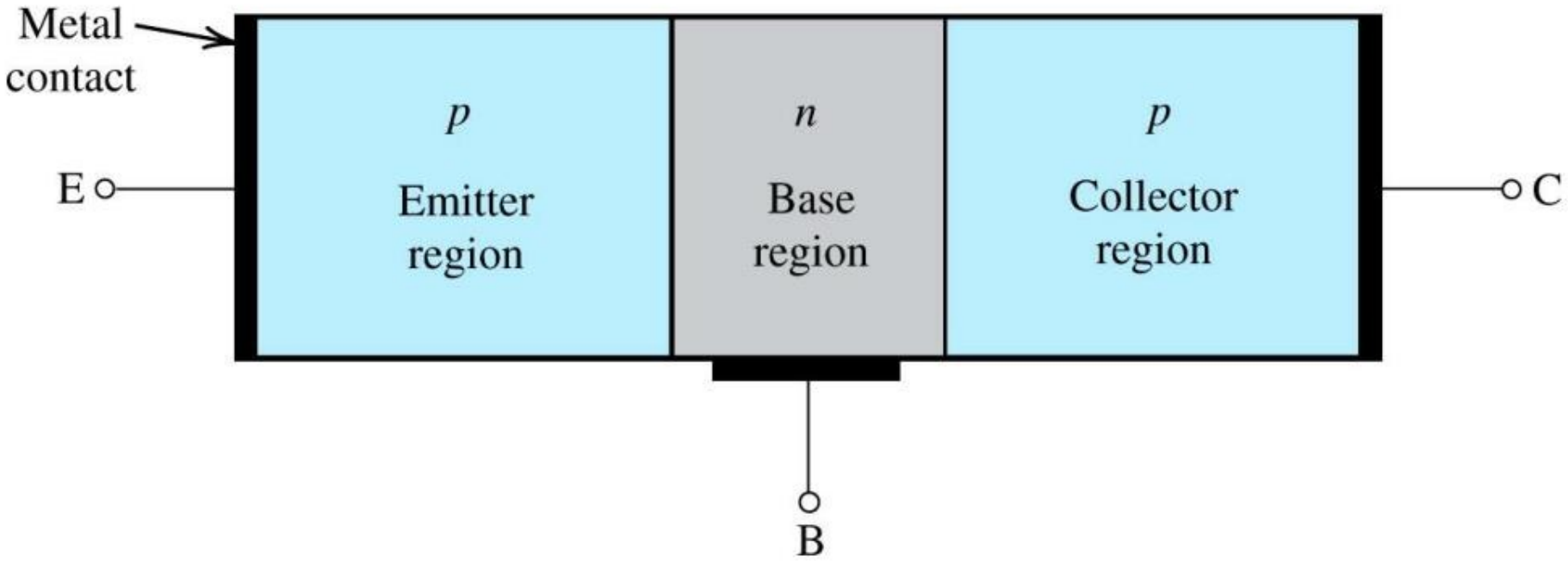


Figure 6.2: A simplified structure of the *pnp* transistor.

Table 6.1 BJT Modes of Operation

Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward

6.1.2. **Operation of the *npn*-Transistor in the Active Mode**

- Active mode is "most important."
- Two external voltage sources are required for biasing to achieve it.
- Refer to Figure 6.3.

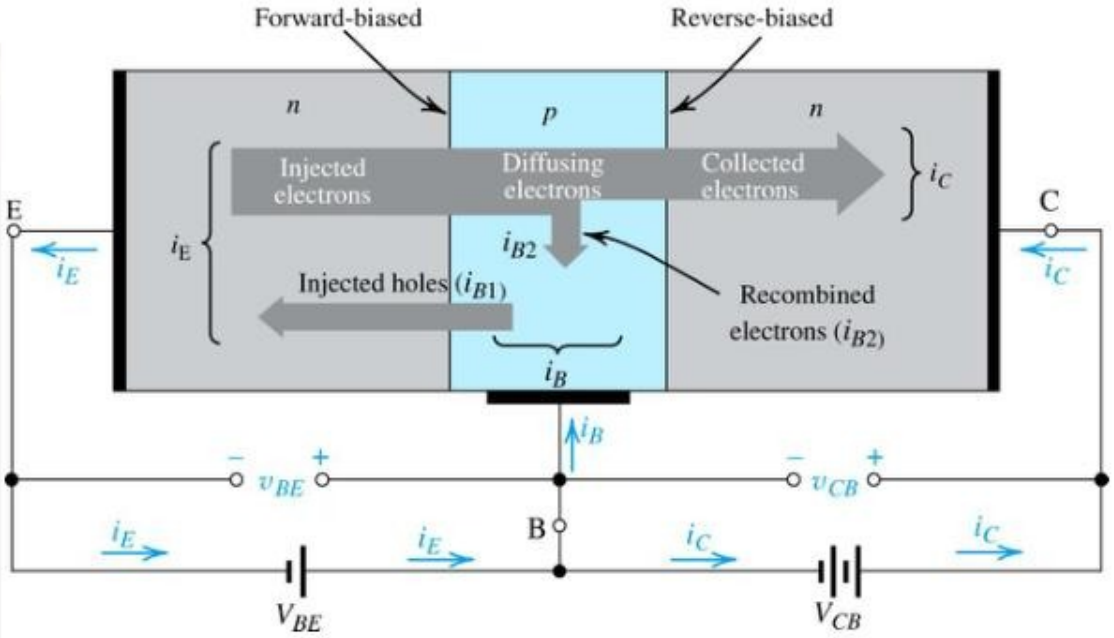


Figure 6.3: Current flow in an *npn* transistor biased to operate in the active mode. (Reverse current components due to drift of thermally generated minority carriers are not shown.)

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Current Flow

- Forward bias on emitter-base junction will cause current to flow.
- This current has two components:
  - electrons injected from emitter into base
  - holes injected from base into emitter.
- It will be shown that first (of the two above) is desirable.
  - This is achieved with heavy doping of emitter, light doping of base.

Current Flow

- **emitter current** (i_E) – is current which flows across EBJ
  - Flows "out" of emitter lead
- **minority carriers** – in *p*-type region.
  - These electrons will be injected from emitter into base.
  - Opposite direction.
- Because base is thin, concentration of excess minority carriers within it will exhibit constant gradient.

$\mathbf{n_p}(x)$=concentration of minority carriers a position x (where 0 represents EBJ boundary)
n_{p0} = thermal-equilibrium value of minority carrier (electron) concentration in base region
v_{BE} =voltage applied across base-emitter junction
V_T =thermal voltage (constant)

(eq6.1) $\mathbf{n_p}(0) = n_{p0} e^{v_{BE}/V_T}$

Straight line represents constant gradient.

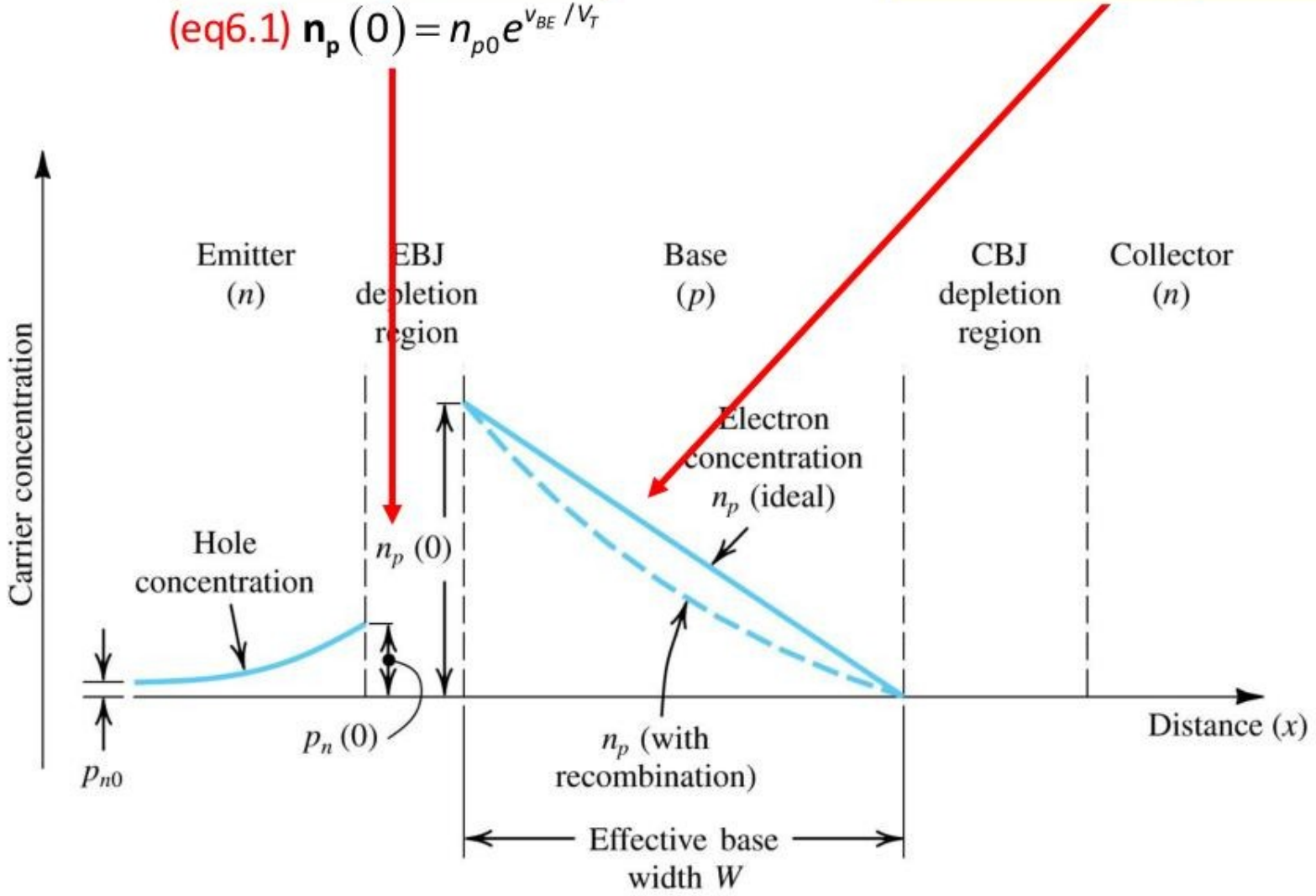


Figure 6.4 Profiles of minority-carrier concentrations in the base and in the emitter of an *npn* transistor operating in the active mode: $v_{BE} > 0$ and $v_{CB} \geq 0$.

Current Flow

- Concentration of minority carrier n_p at boundary EBJ is defined by (6.1).
- Concentration of minority carriers n_p at boundary of CBJ is zero.
  - Positive v_{CB} causes these electrons to be swept across junction.

$\mathbf{n_p}(x)$=concentration of minority carriers a position…
…x (where 0 represents EBJ boundary)
n_{p0}= thermal-equilibrium value of minority carrier…
…(electron) concentration in base region
v_{BE} =voltage applied across base-emitter junction
V_T =thermal voltage (constant)

$$(\text{eq6.1})\ \mathbf{n_p}(0) = n_{p0} e^{v_{BE}/V_T}$$

Current Flow

- Tapered minority-carrier concentration profile exists.
- It causes electrons injected into base to diffuse through base toward collector.
- As such, electron diffusion current (I_n) exists.

A_E =cross-sectiona area of the base-emitter junction
q= magnitude of the electron charge
D_n= electron diffusivity in base
W= width of base

$$\text{(eq6.2)}\ I_n = A_E q D_n \frac{dn_p(x)}{dx}$$

$$\text{(eq6.2)}\ I_n = A_E q D_n \left(\frac{-dn_p(0)}{W} \right)$$

this simplification may be made if gradient assumed to be straight line

Current Flow

- Some "diffusing" electrons will combine with holes (majority carriers in base).
- Base is thin, however, and recombination is minimal.
- Recombination does, however, cause gradient to take slightly curved shape.
  - The straight line is assumed.

$\mathbf{n_p}(x)$=concentration of minority carriers a position x (where 0 represents EBJ boundary)
n_{p0} = thermal-equilibrium value of minority carrier (electron) concentration in base region
v_{BE} =voltage applied across base-emitter junction
V_T =thermal voltage (constant)

(eq6.1) $\mathbf{n_p}(0) = n_{p0} e^{v_{BE}/V_T}$

Recombination causes actual gradient to be curved, not straight.

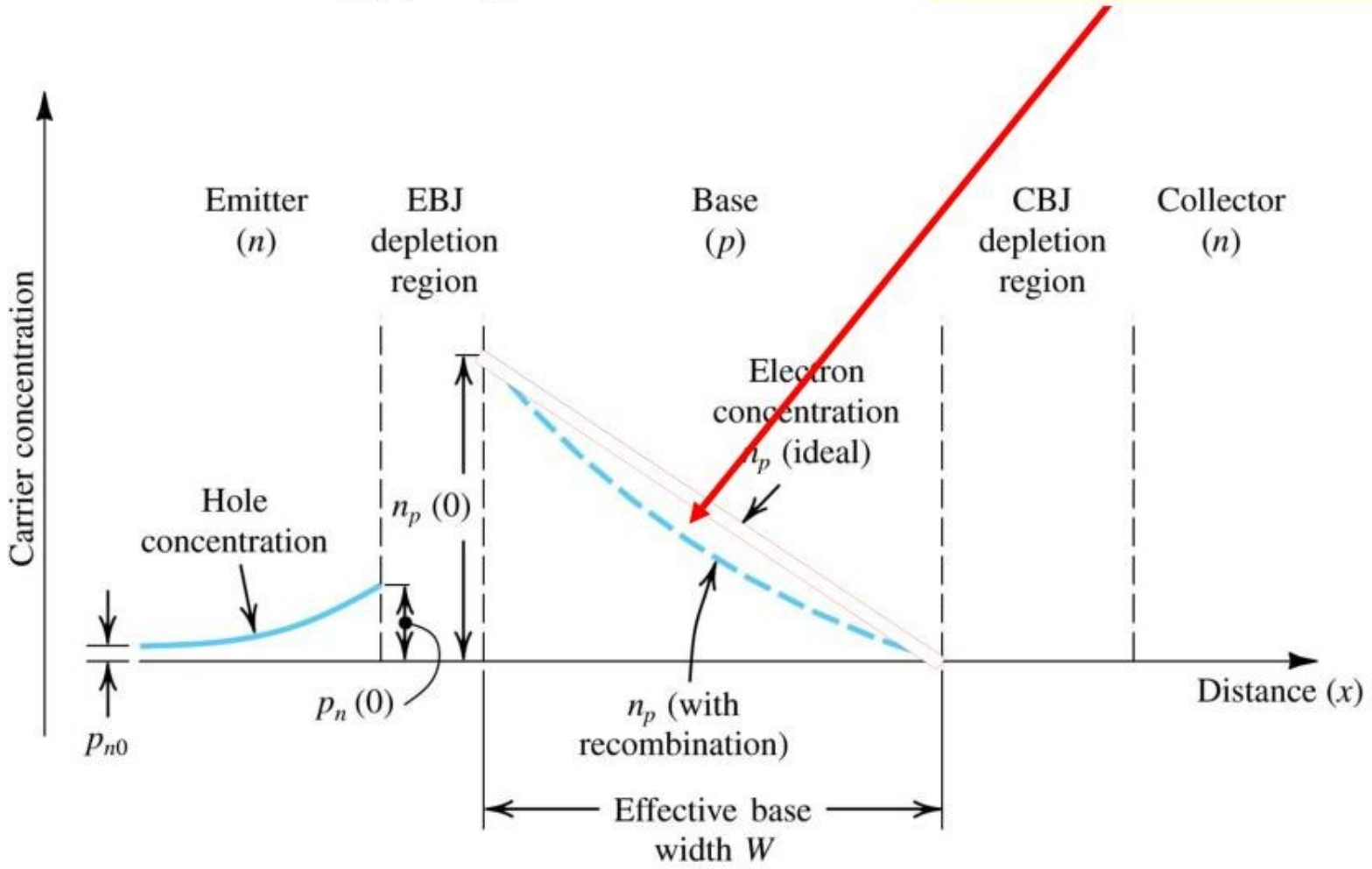


Figure 6.4 Profiles of minority-carrier concentrations in the base and in the emitter of an *npn* transistor operating in the active mode: $v_{BE} > 0$ and $v_{CB} \geq 0$.

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The Collector Current

- It is observed that most diffusing electrons will reach boundary of collector-base depletion region.
- Because collector is more positive than base, these electrons are swept into collector.
  - **collector current** (i_C) is approximately equal to I_n.
  - $i_C = I_n$

(eq6.3) $i_C = I_S \mathbf{e}^{v_{BE}/V_T}$

saturation current: $I_S = \dfrac{A_E q D_n n_{p0}}{W}$

(eq6.4) $I_S = \dfrac{A_E q D_n}{W} \dfrac{n_i^2}{N_A}$

ni= intrinsic carrier density
NA= doping concentration of base

The Collector Current

- Magnitude of i_C is independent of v_{CB}.
  - As long as collector is positive, with respect to base.
- **saturation current** (I_S) – is inversely proportional to W and directly proportional to area of EBJ.
  - Typically between 10^{-12} and $10^{-18}A$
  - Also referred to as **scale current**.

The Base Current

- **base current** (i_B) – composed of two components:
  - i_{b1} – due to holes injected from base region into emitter.
  - i_{b2} – due to holes that have to be supplied by external circuit to replace those recombined.

β= transistor parameter

$$\text{(eq6.5)}\quad i_B = \frac{i_C}{\beta}$$

$$\text{(eq6.6)}\quad i_B = \frac{I_S}{\beta}\mathbf{e}^{v_{BE}/V_T}$$

The Base Current

- **common-emitter current gain** (β) – is influenced by two factors:
  - width of base region (W)
  - relative doping of base emitter regions (N_A/N_D)
- High Value of β
  - thin base (small W in nano-meters)
  - lightly doped base / heavily doped emitter (small N_A/N_D)

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The Emitter Current

- All current which enters transistor must leave.
  - $i_E = i_C + i_B$
- Equations (6.7) through (6.13) expand upon this idea.

this expression is generated through combination of (6.5) and (6.7)

(eq6.8/6.9) $$i_E = \frac{\beta+1}{\beta} i_C = \frac{\beta+1}{\beta} \underbrace{\left(I_S \mathbf{e}^{v_{BE}/V_T} \right)}_{i_C}$$

(eq6.10) $$i_C = \alpha i_E$$

this parameter is reffered to as **common-base current gain**

(eq6.11) $$\alpha = \frac{\beta}{\beta+1}$$, (eq6.13) $$\beta = \frac{\alpha}{1-\alpha}$$

(eq6.12) $$i_E = \frac{I_S}{\alpha} \mathbf{e}^{v_{BE}/V_T}$$

Recapitulation and Equivalent-Circuit Models

- Previous slides present first-order BJT model.
  - Assumes *npn* transistor in active mode.
- Basic relationship is collector current (i_C) is related exponentially to forward-bias voltage (v_{BE}).
  - It remains independent of v_{CB} as long as this junction remains reverse biased.
    - $v_{CB} > 0$

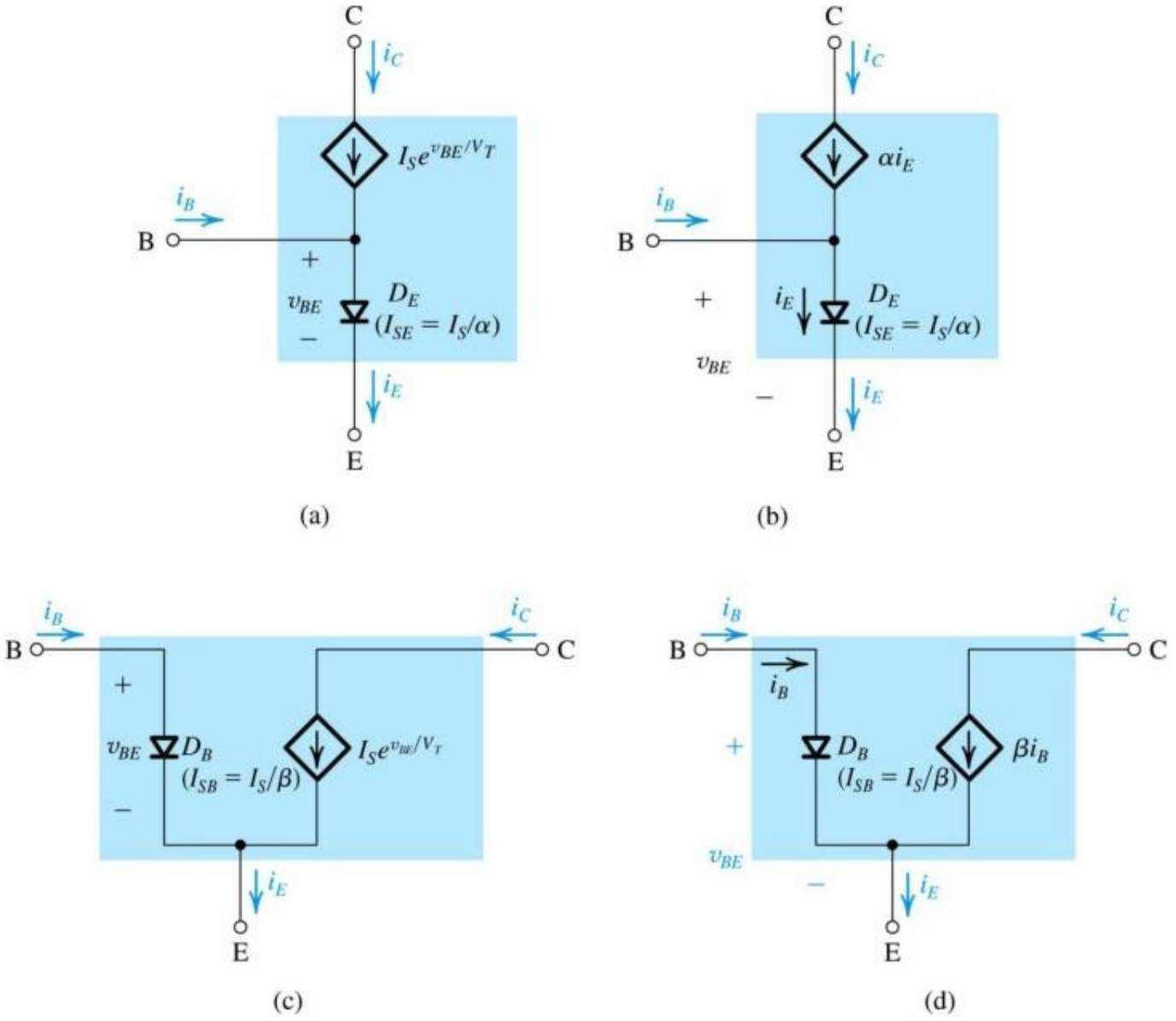


Figure 6.5: Large-signal equivalent-circuit models of the *npn* BJT operating in the forward active mode.

Example 6.1.

- Refer to textbook for Example 6.1.

6.1.3. **Structure of Actual Transistors**

- Figure 6.7 shows a more realistic BJT cross-section.
- Collector virtually surrounds entire emitter region.
  - This makes it difficult for electrons injected into base to escape collection.
- Device is not symmetrical.
  - As such, emitter and collector cannot be interchanged.
  - Device is uni-directional.

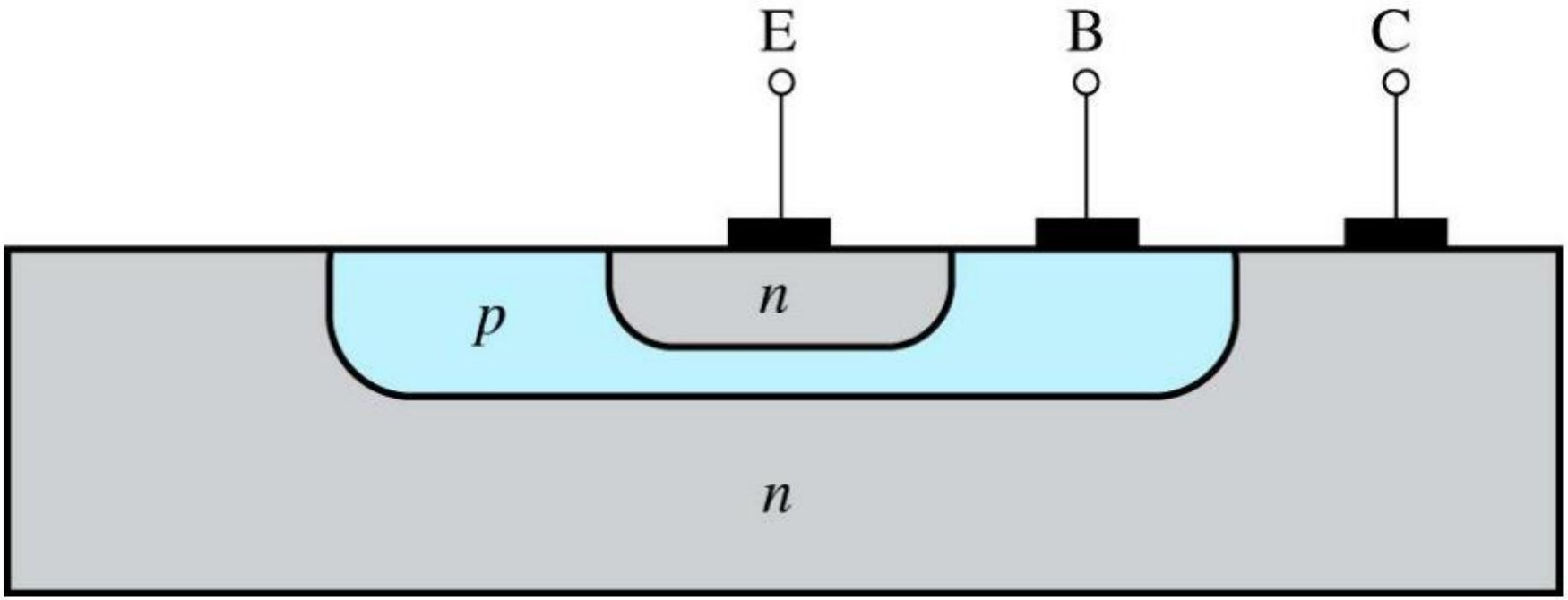


Figure 6.7: Cross-section of an *npn* BJT.

6.1.4. **Operation in Saturation Mode**

- For BJT to operate in active mode, CBJ must be reverse biased.
  - However, for small values of forward-bias, a *pn*-junction does not operate effectively.
- As such, active mode operation of *npn*-transistor may be maintained for v_{CB} down to approximately -0.4V.
  - Only after this point will "diode" begin to really conduct.

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6.1.4. Operation in Saturation Mode

(eq6.14) collector current in saturation region:

$$i_C = I_S e^{v_{BE}/V_T} - \overbrace{I_{SC} e^{v_{BC}/V_T}}^{I_{SC} =}$$

this terms plays bigger role as v_{BC} exceeds 0.4V

(eq6.15) base current in saturation region:

$$i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T} + I_{SC} e^{v_{BC}/V_T}$$

(eq6.16) forced β:

$$\beta_{forced} = \left.\frac{i_C}{i_B}\right|_{saturation} \leq \beta$$

As v_{BC} is increased, the value of β is forced lower and lower.

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6.1.4. **Operation in Saturation Mode**

- Two questions must be asked to determine whether BJT is in saturation mode, or not:
  - Is the CBJ forward-biased by more than $0.4V$?
  - Is the ratio i_C/i_B less than β?

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6.1.5. The *pnp* Transistor

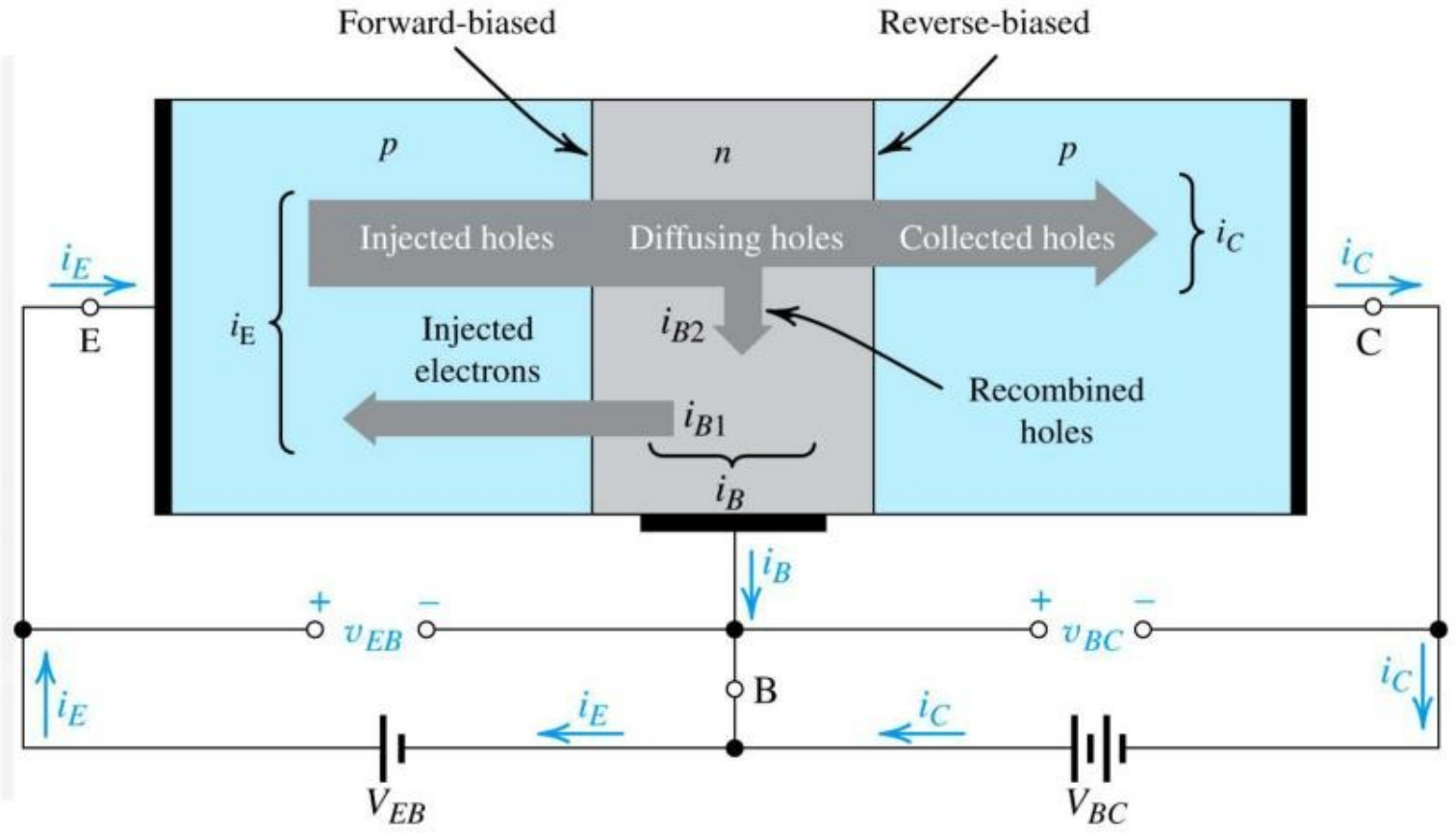


Figure 6.10: Current flow in a *pnp* transistor biased to operate in the active mode.

6.1.5. The *pnp* Transistor

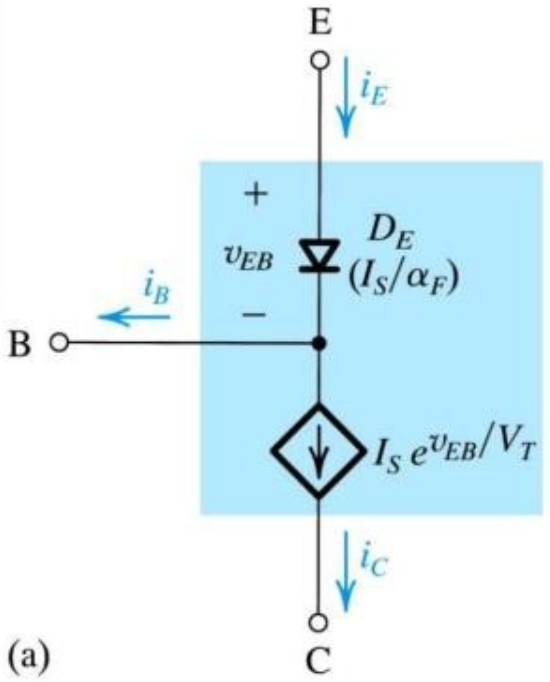
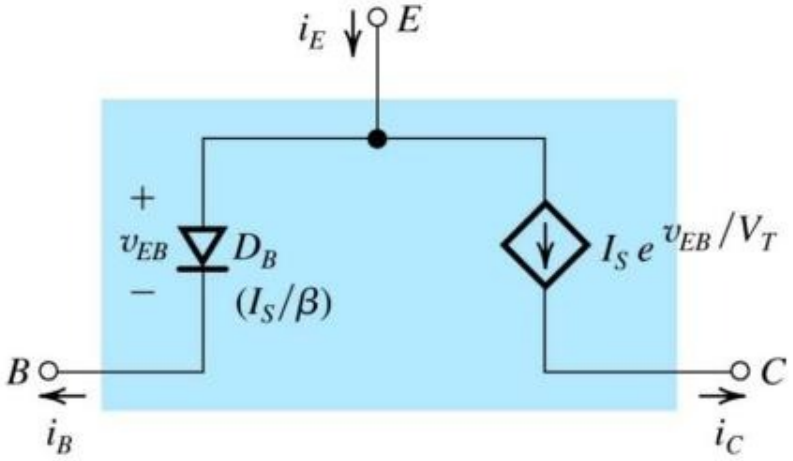


Figure 6.11: Two large-signal models for the *pnp* transistor operating in the active mode.



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6.2. Current-Voltage Characteristics

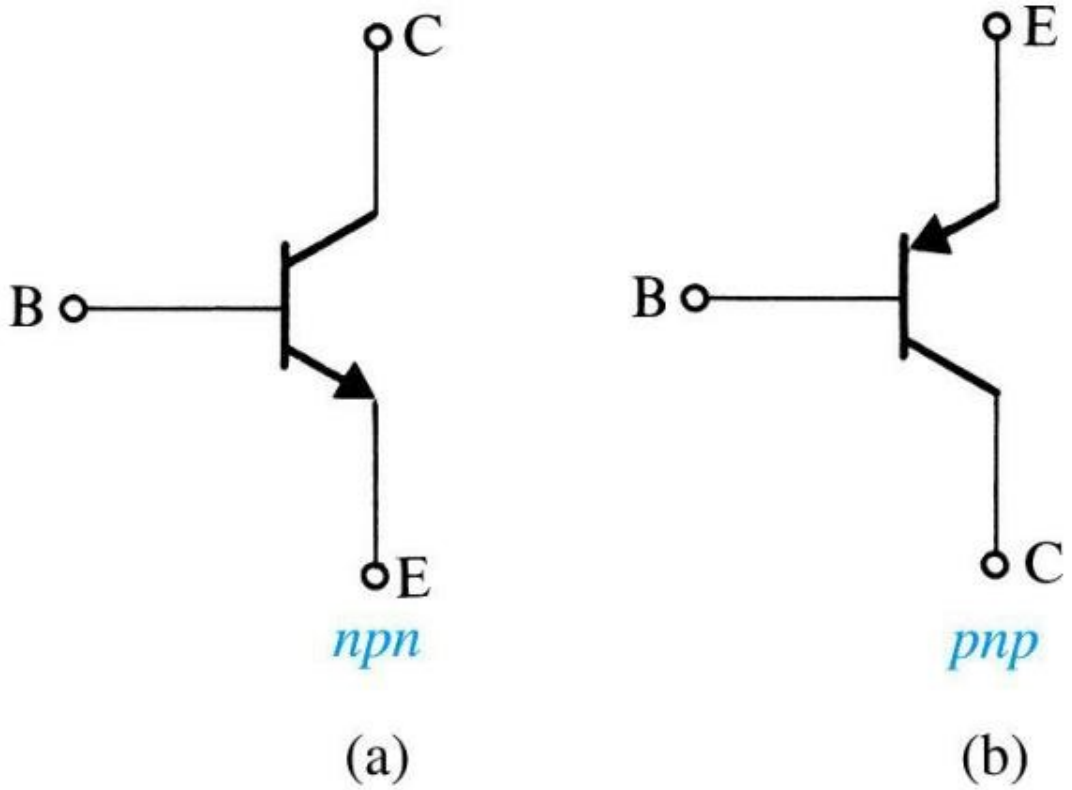


Figure 6.12: Circuit symbols for BJTs.

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6.2.1. Circuit Symbols and Conventions

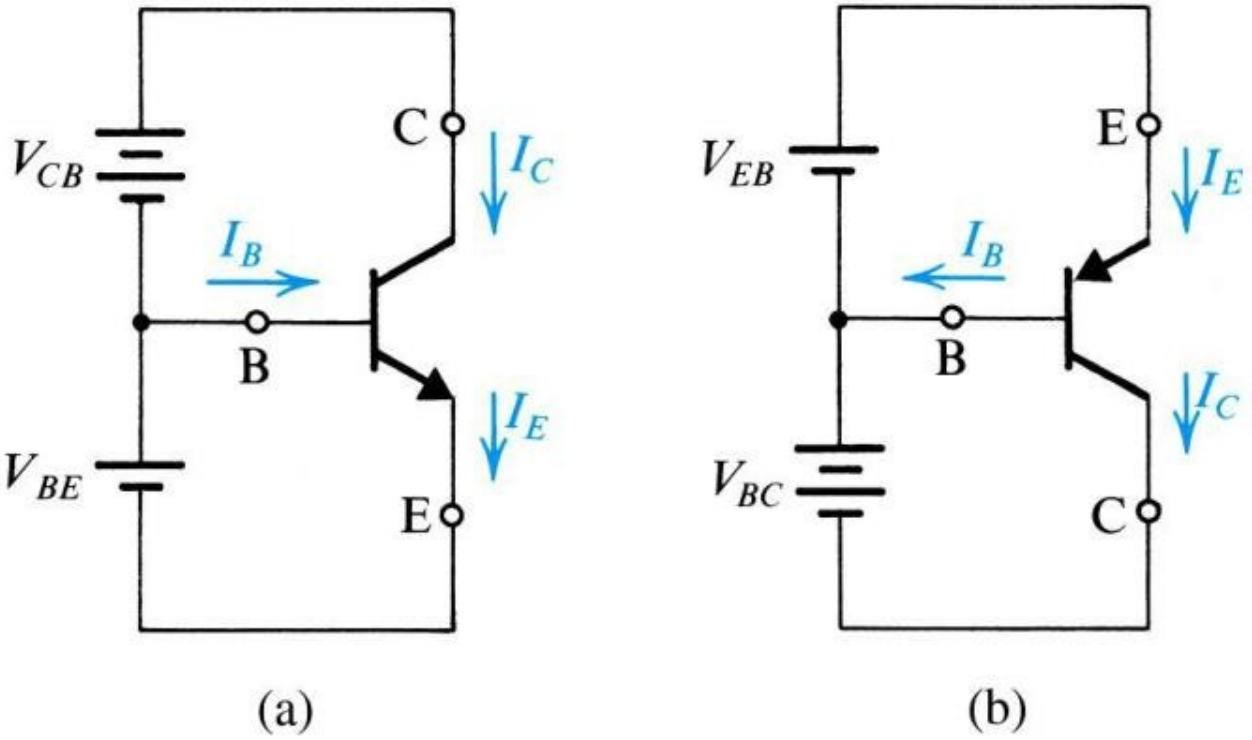


Figure 6.13: Voltage polarities and current flow in transistors biased in the active mode.

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6.2.1. Circuit Symbols and Conventions

Table 6.2 Summary of the BJT Current–Voltage Relationships in the Active Mode

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the *pnp* transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E \qquad i_B = (1-\alpha) i_E = \frac{i_E}{\beta+1}$$

$$i_C = \beta i_B \qquad i_E = (\beta+1) i_B$$

$$\beta = \frac{\alpha}{1-\alpha} \qquad \alpha = \frac{\beta}{\beta+1}$$

V_T = thermal voltage $= \dfrac{kT}{q} \simeq 25$ mV at room temperature

The Collector-Base Reverse Current (I_{CB0})

- Previously, small reverse current was ignored.
  - This is carried by thermally-generated minority carriers.
- However, it does deserve to be addressed.
- The **collector-base junction current** (I_{CBO}) is normally in the nano-ampere range.
  - Many times higher than its theoretically-predicted value.

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6.2.2. Graphical Representation of Transistor Characteristics

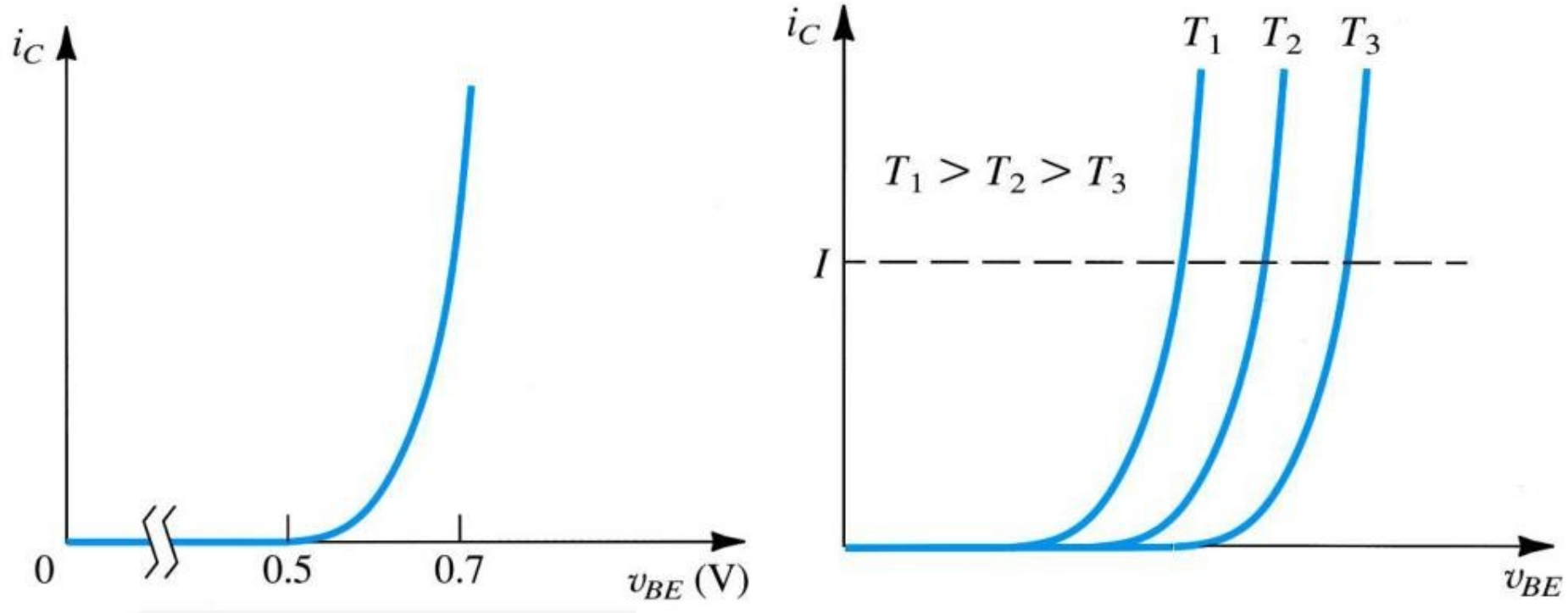


Figure 6.15/16: (left) The i_C-v_{BE} characteristic for an npn transistor. **(right)** Effect of temperature on the i_C-v_{BE} characteristic. Voltage polarities and current flow in transistors biased in the active mode.

6.2.3. Dependence of i_C on Collector Voltage – The Early Effect

- When operated in active region, practical BJT's show some dependence of collector current on collector voltage.
- As such, i_C-v_{CB} characteristic is not "straight".

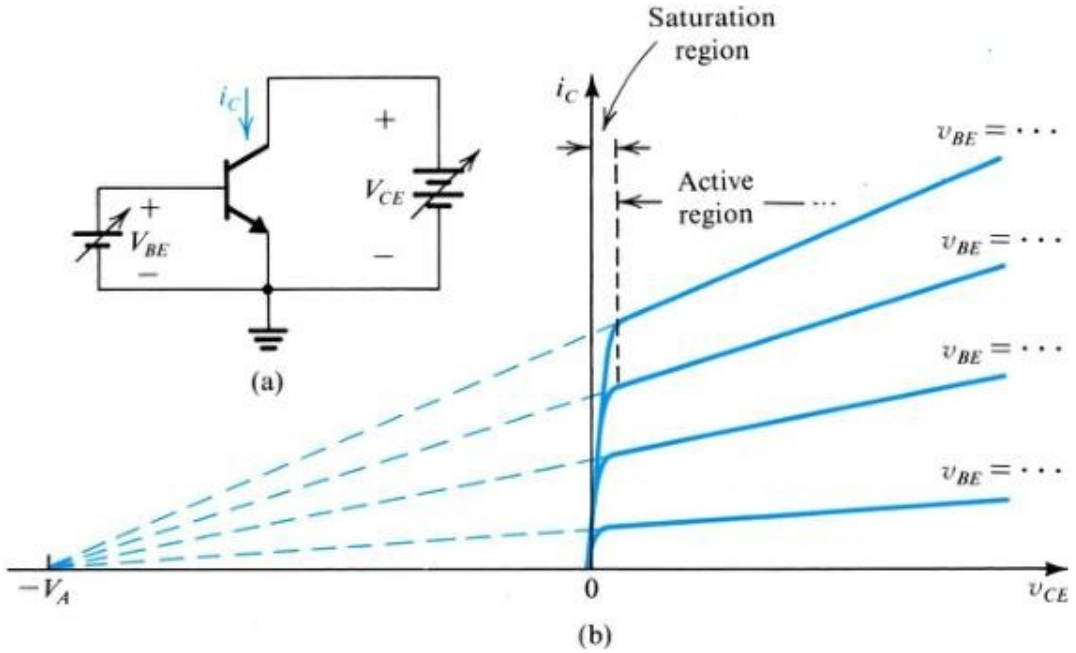


Figure 6.17 (a) Conceptual circuit for measuring the i_C–v_{CE} characteristics of the BJT. **(b)** The i_C–v_{CE} characteristics of a practical BJT.

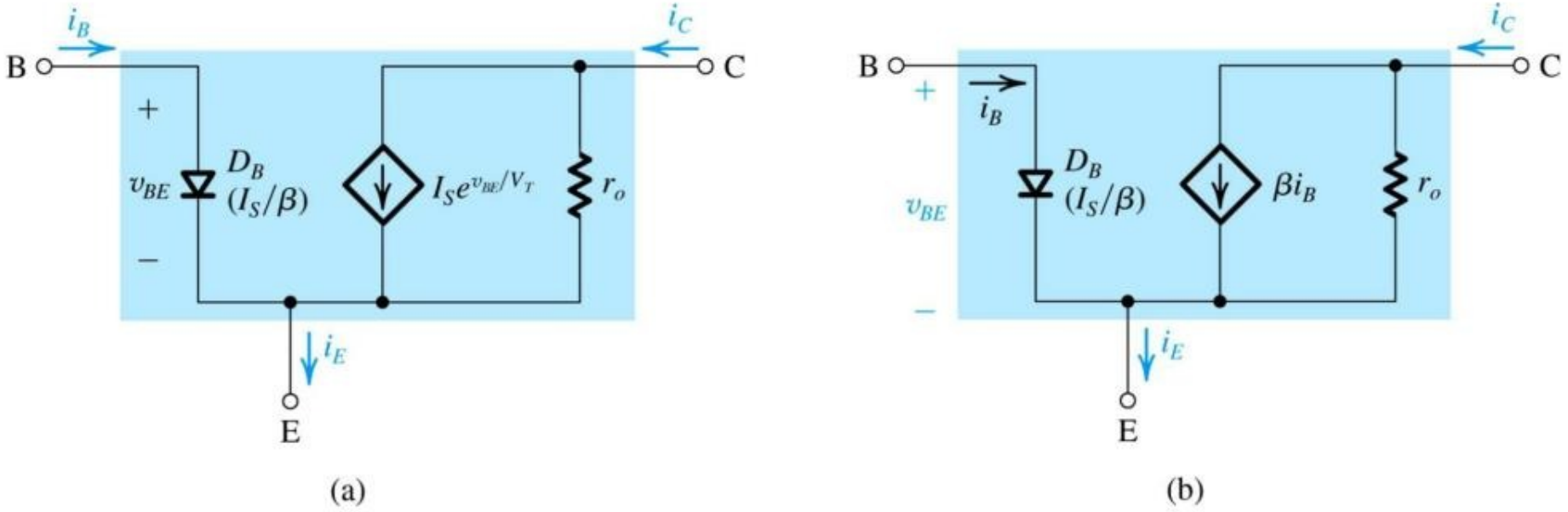
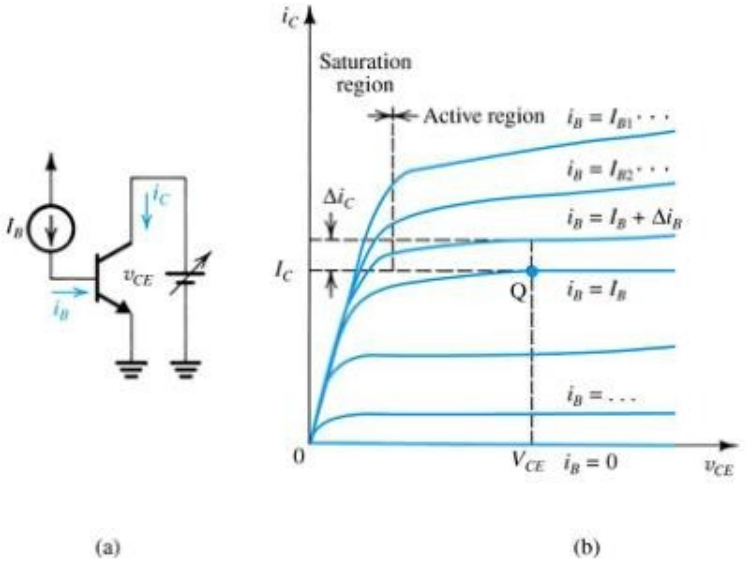


Figure 6.18: Large-signal equivalent-circuit models of an *npn* BJT operating in the active mode in the common-emitter configuration with the output resistance r_o included.

6.2.4. An Alternative Form of the Common-Emitter Characteristics

- The Common-Emitter Current Gain
  - A second way to quantify β is changing base current by Δi_B and measuing incremental Δi_C.
- The Saturation Voltage V_{CEsat} and Saturation Resistance



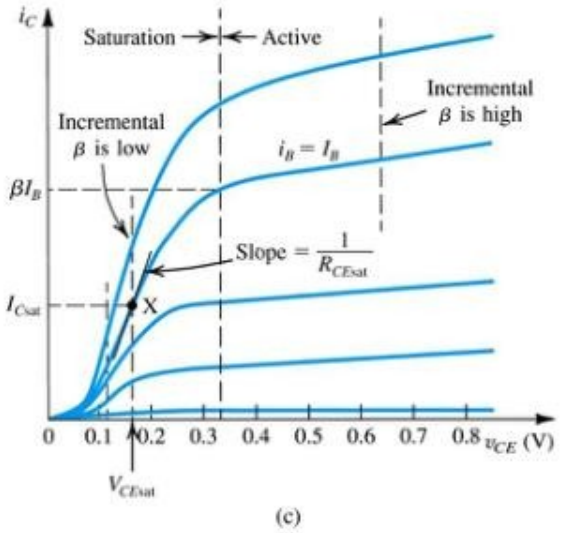


Figure 6.19: Common-emitter characteristics. (**a**) Basic CE circuit; note that in (**b**) the horizontal scale is expanded around the origin to show the saturation region in some detail. A much greater expansion of the saturation region is shown in (**c**).

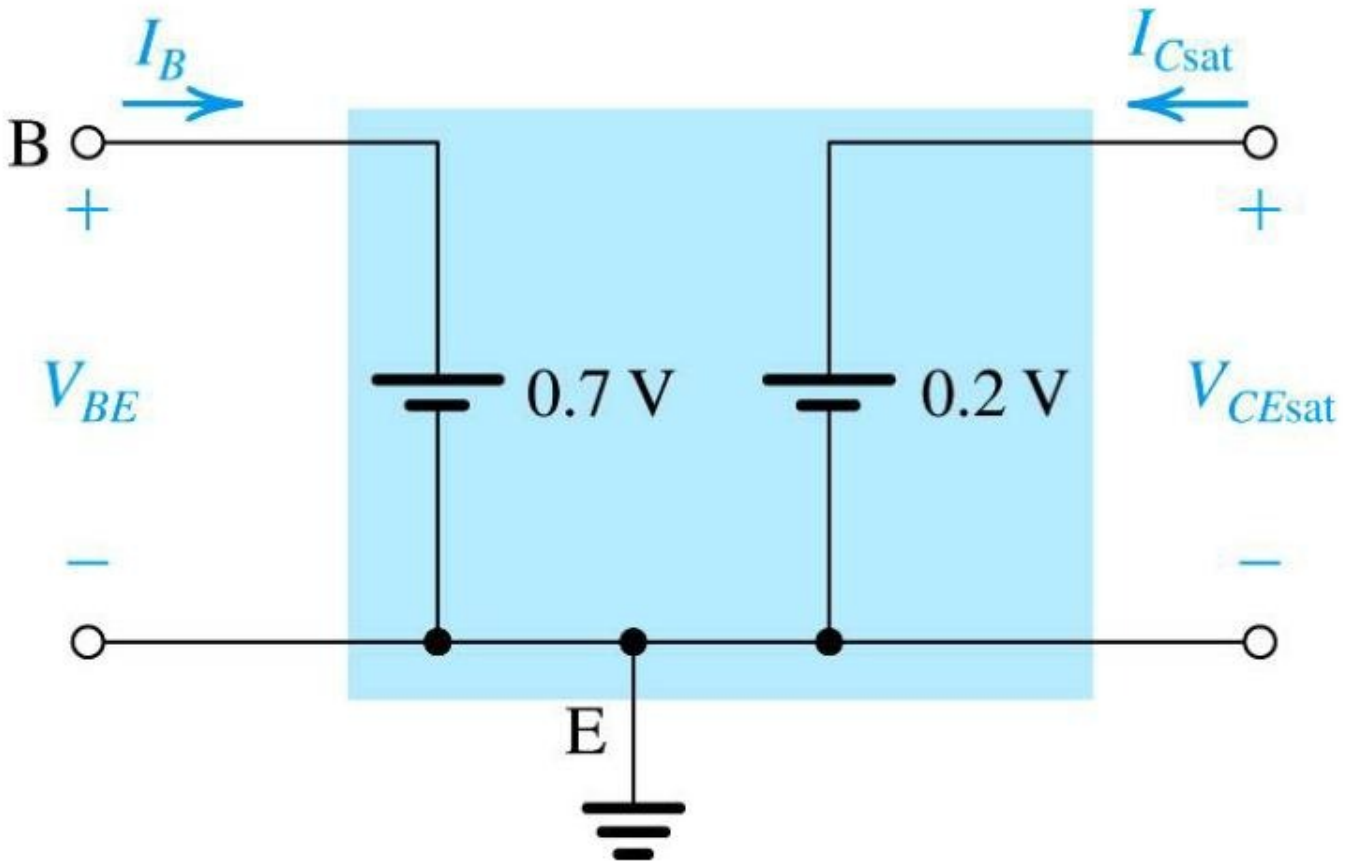
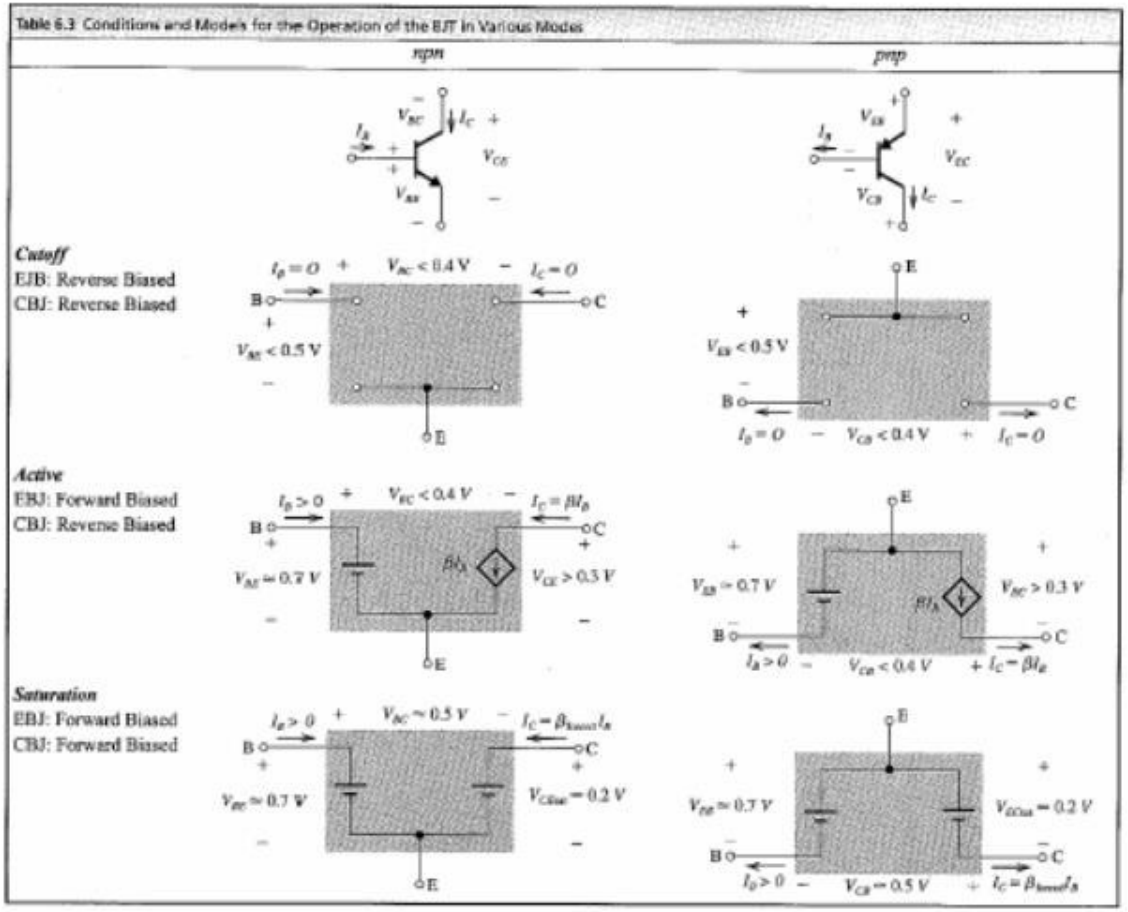


Figure 6.20: A simplified equivalent-circuit model of the saturated transistor.

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6.3. **BJT Circuits at DC**

Table 6.3 Conditions and Models for the Operation of the BJT in Various Modes



6.4. Applying the BJT in Amplifier Design

- Similar to the configuration presented in Chapter 5, an amplifier may be designed by transistor and series resistance.
- However, it is necessary to model the **voltage transfer characteristic** (VTC).
  - Equation (6.26)
- Appropriate biasing is important to ensure linear gain, and appropriate input voltage swing.
  - Small-signal model is employed to model the amp's operation.

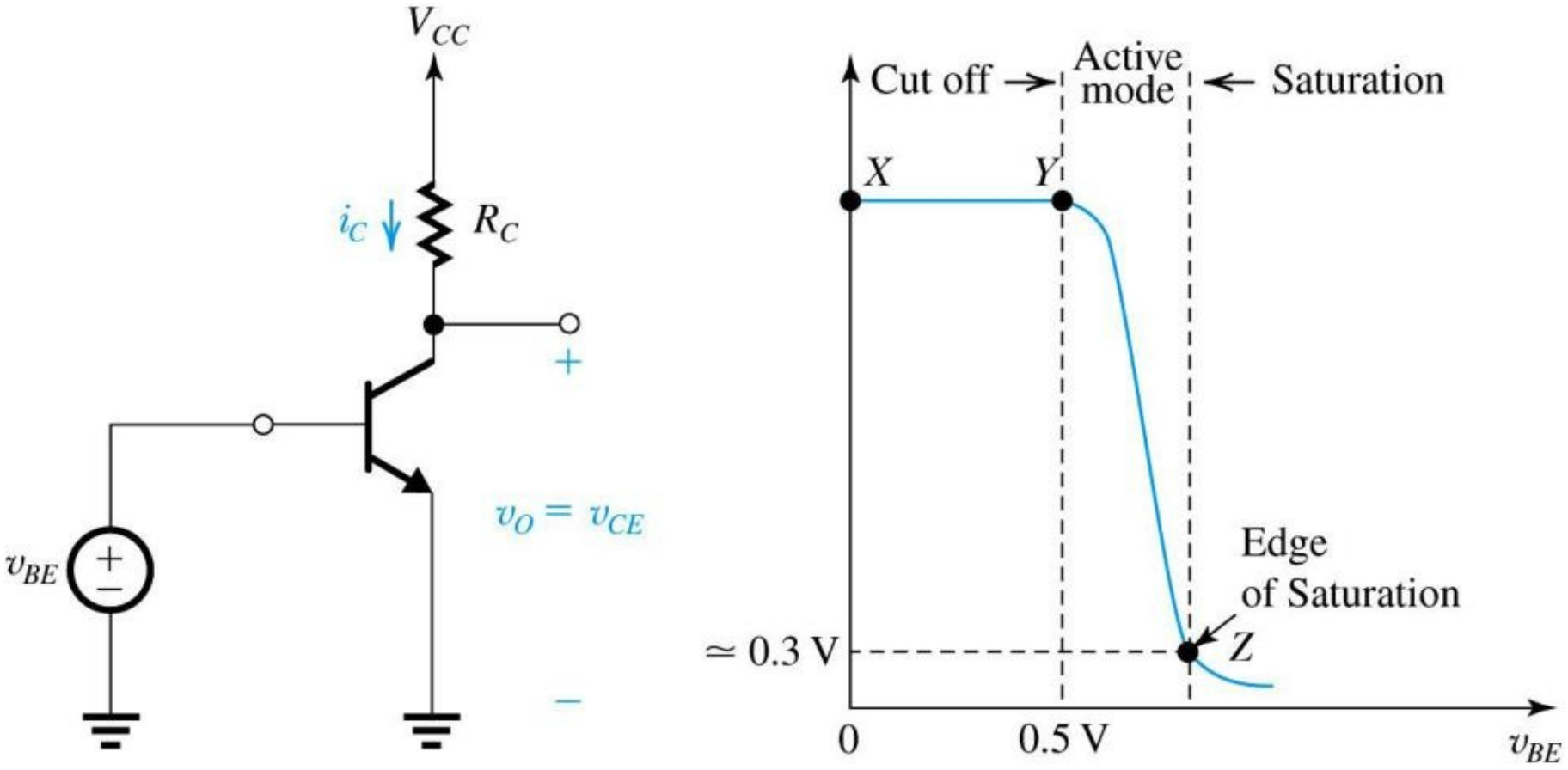


Figure 6.31 (a) Simple BJT amplifier with input v_{BE} and output v_{CE}. **(b)** The voltage transfer characteristic (VTC) of the amplifier in **(a)**. The three segments of the VTC correspond to the three modes of operation of the BJT.

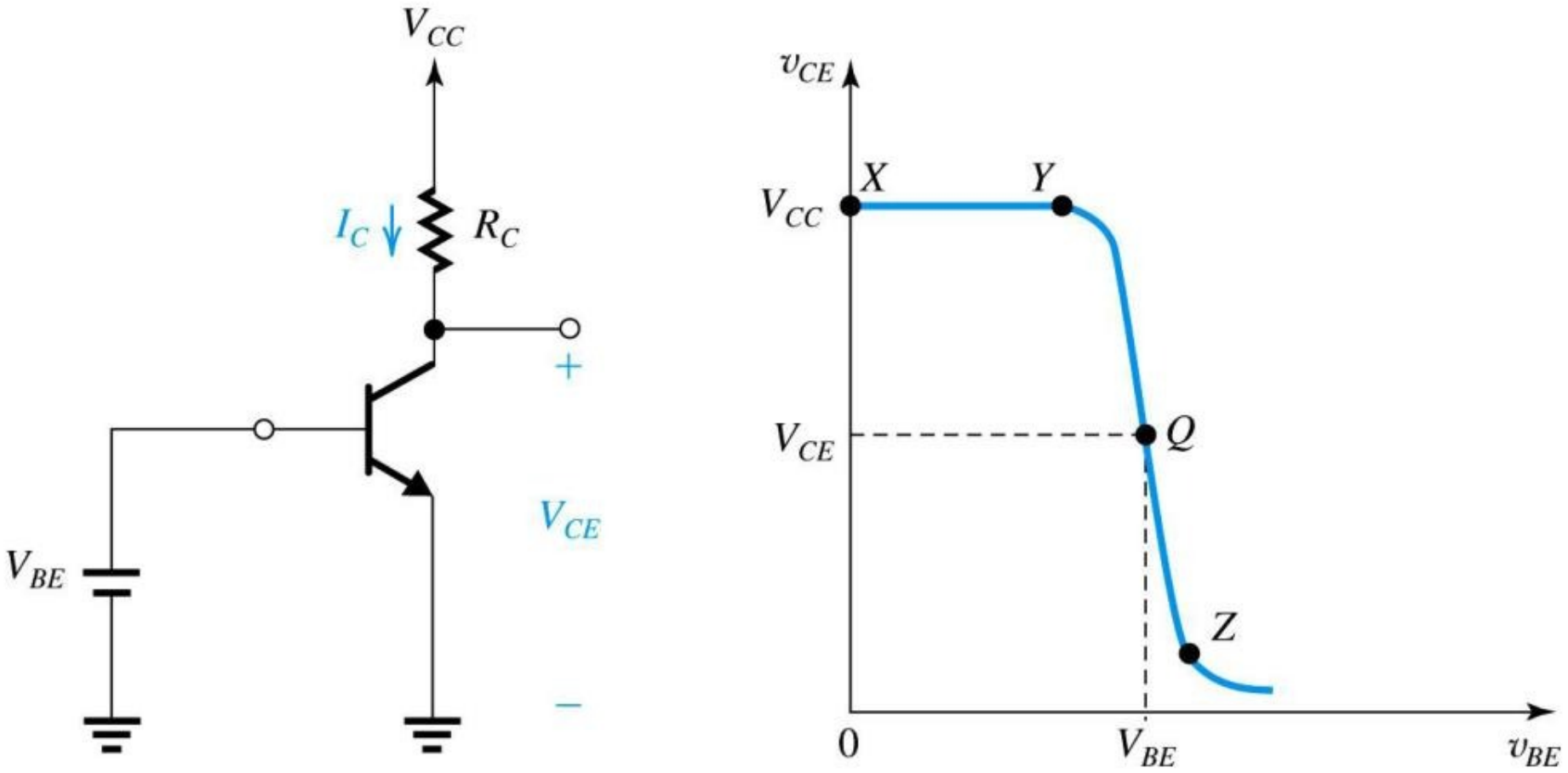


Figure 6.32: Biasing the BJT amplifier at a point Q located on the active-mode segment of the VTC.

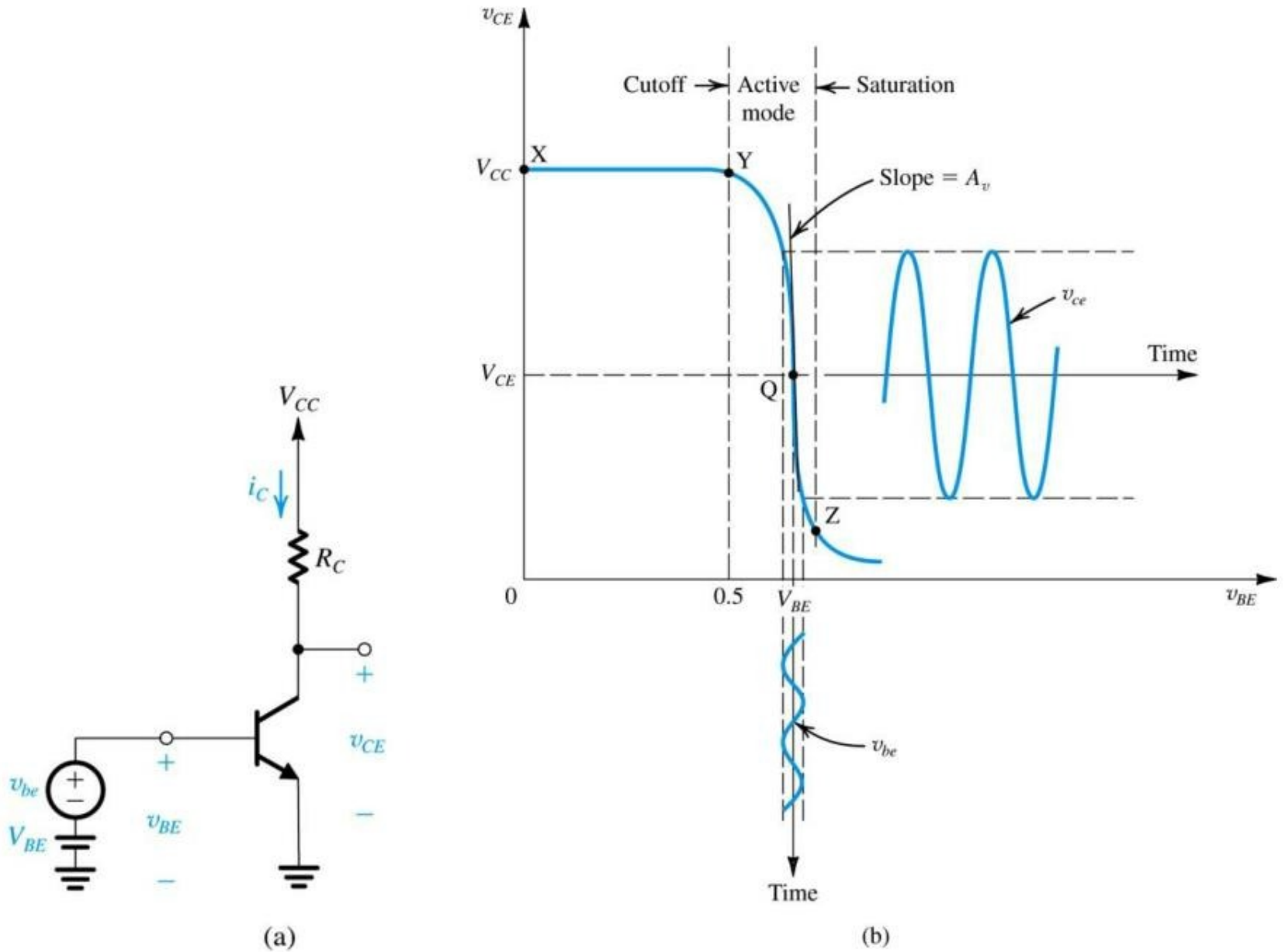


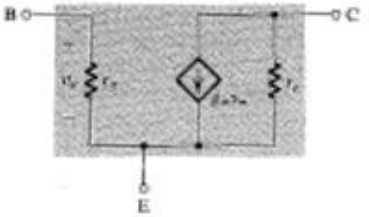
Figure 6.33 BJT amplifier biased at a point Q, with a small voltage signal v_{be} superimposed on the dc bias voltage V_{BE}. The resulting output signal v_{ce} appears superimposed on the dc collector voltage V_{CE}. The amplitude of v_{ce} is larger than that of v_{be} by the voltage gain A_v.

6.6. Basic BJT Amplifier Configurations

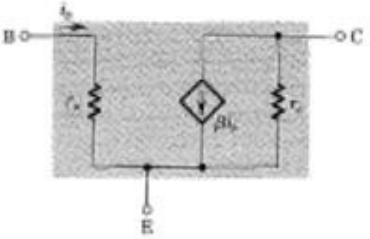
Table 6.4 Small-Signal Models of the BJT

Hybrid-π Model

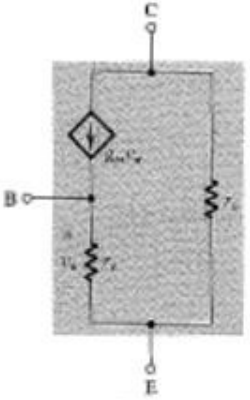
■ $(g_m v_\pi)$ Version



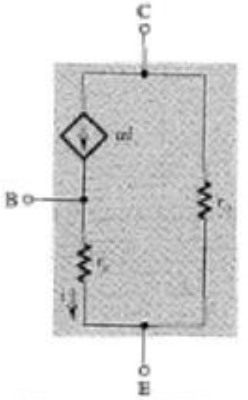
■ (βi_b) Version



■ $(g_m v_\pi)$ Version



■ (αi) Version



Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \qquad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \qquad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \qquad r_o = \frac{|V_A|}{I_C}$$

In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

In Terms of r_e

$$g_m = \frac{\alpha}{r_e} \qquad r_\pi = (\beta+1)r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

Relationships between α and β

$$\beta = \frac{\alpha}{1-\alpha} \qquad \alpha = \frac{\beta}{\beta+1} \qquad \beta+1 = \frac{1}{1-\alpha}$$

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6.6.1. Three-Basic Configurations

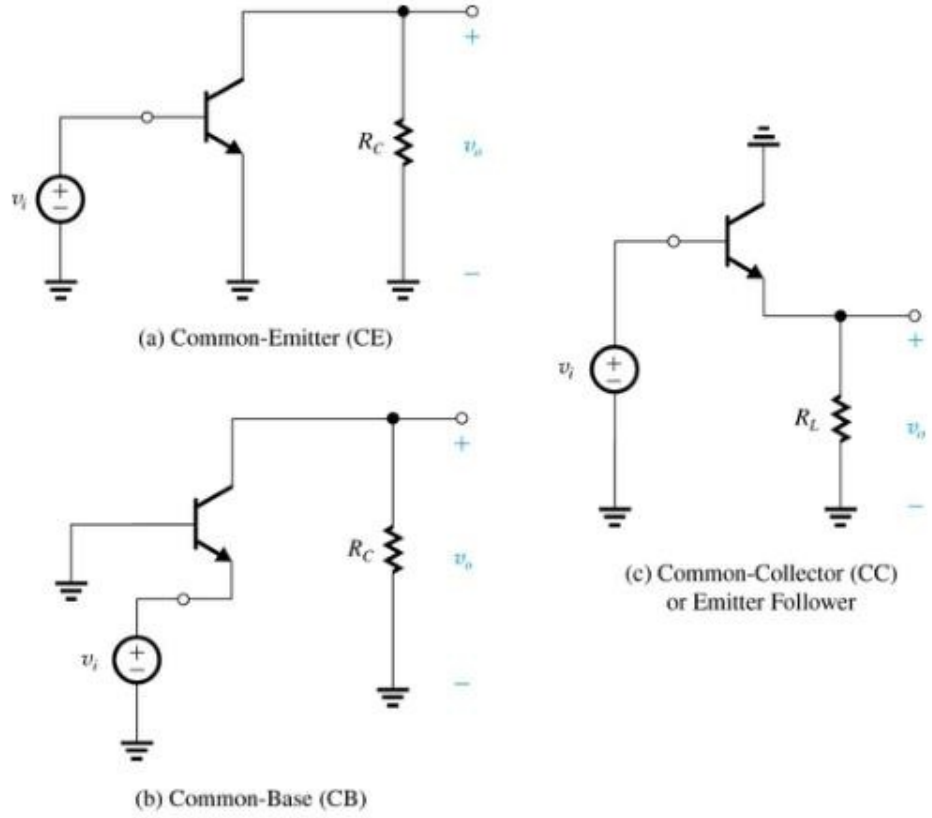


Figure 6.48: The three basic configurations of BJT amplifier. The biasing arrangements are not shown.

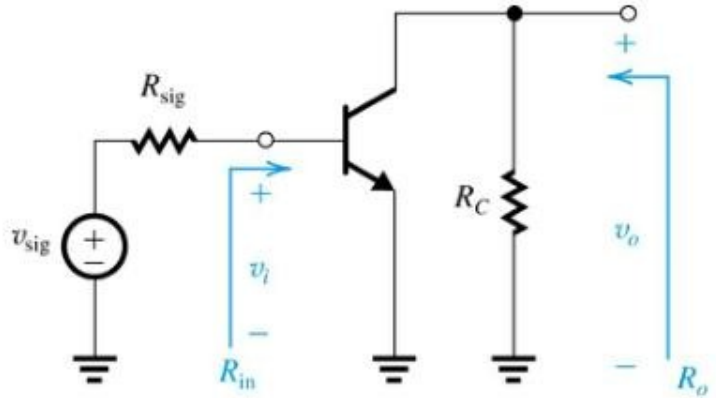
6.6.3. The Common-Emitter (CE) Amplifier

- Of three configurations, the CE amplifier is most widely used.
- Figure 6.50(a) shows a common-emitter amplifier – with biasing arrangement omitted.
  - signal course (v_{sig})
  - source resistance (R_{sig})
  - input resistance (R_{in})
  - gain (A_{vo})
  - output resistance (R_o)
  - transconductance (G_v)

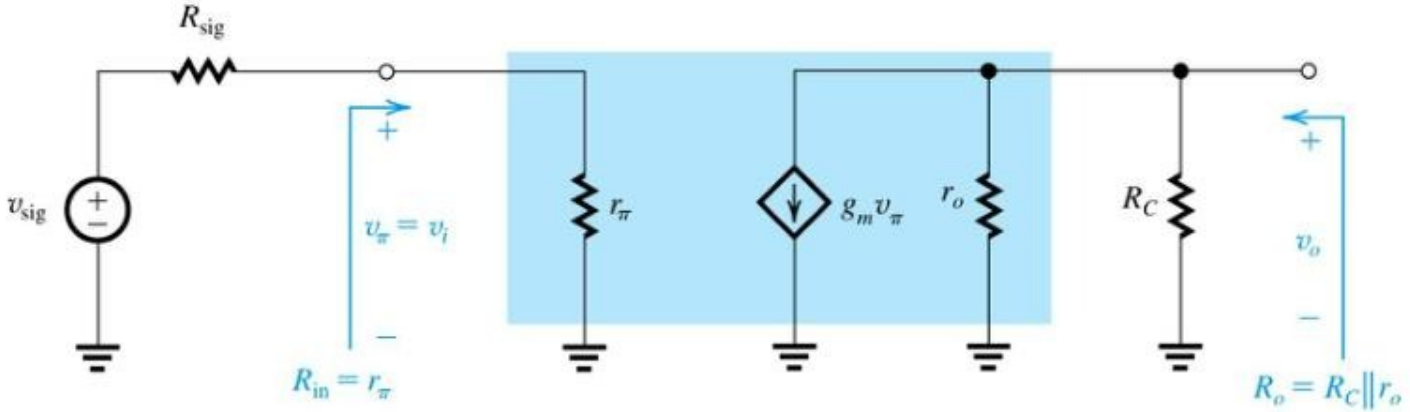
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Common-Emitter Amplifier

Figure 6.50: (a) Common-Emitter Amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig}. **(b)** The common-emitter amplifier circuit with the BJT replaced with its hybrid-pi model.



(a)



(b)

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Characteristic Parameters of the CE Amplifier

- Replacing BJT with hybrid-pi model yields the expressions to right…

(eq6.69) input resistance: $R_{in} = r_\pi$

---

output voltage: $v_o = -(g_m v_\pi)(R_C \,||\, r_o)$

(eq6.70) open-circuit voltage gain: $A_{vo} = -g_m (R_C \,||\, r_o)$

(eq6.71) oper-circuit voltage gain: $A_{vo} = -g_m R_C$

with r_o neglected

(eq6.72) output resistance: $A_{vo} = -g_m R_C$

Characteristic Parameters of the CE Amplifier

- **Three Observations**
  - The input resistance $R_{in} = r_\pi = \beta/g_m$ is moderate to low in value.
  - The output resistance $R_o = R_C$ is moderate to high in value.
  - The open-circuit voltage gain (A_{vo}) can be high – making the CE configuration the workhorse in BJT amplifier design.

Overall Voltage Gain

(eq6.74) amplifier input voltage: $R_{in} v_i = v_{sig} \dfrac{r_\pi}{r_\pi + R_{sig}} = r_\pi$

(eq6.75) voltage gain: $A_v = -g_m \left(R_C \,||\, R_L \,||\, r_o \right)$

not open-loop

(eq6.76) overall voltage gain: $G_v = \dfrac{v_o}{v_{sig}} = \dfrac{-r_\pi}{r_\pi + R_{sig}} g_m \left(R_C \,||\, R_L \,||\, r_o \right)$

6.6.5. The Common-Base (CB) Amplifier

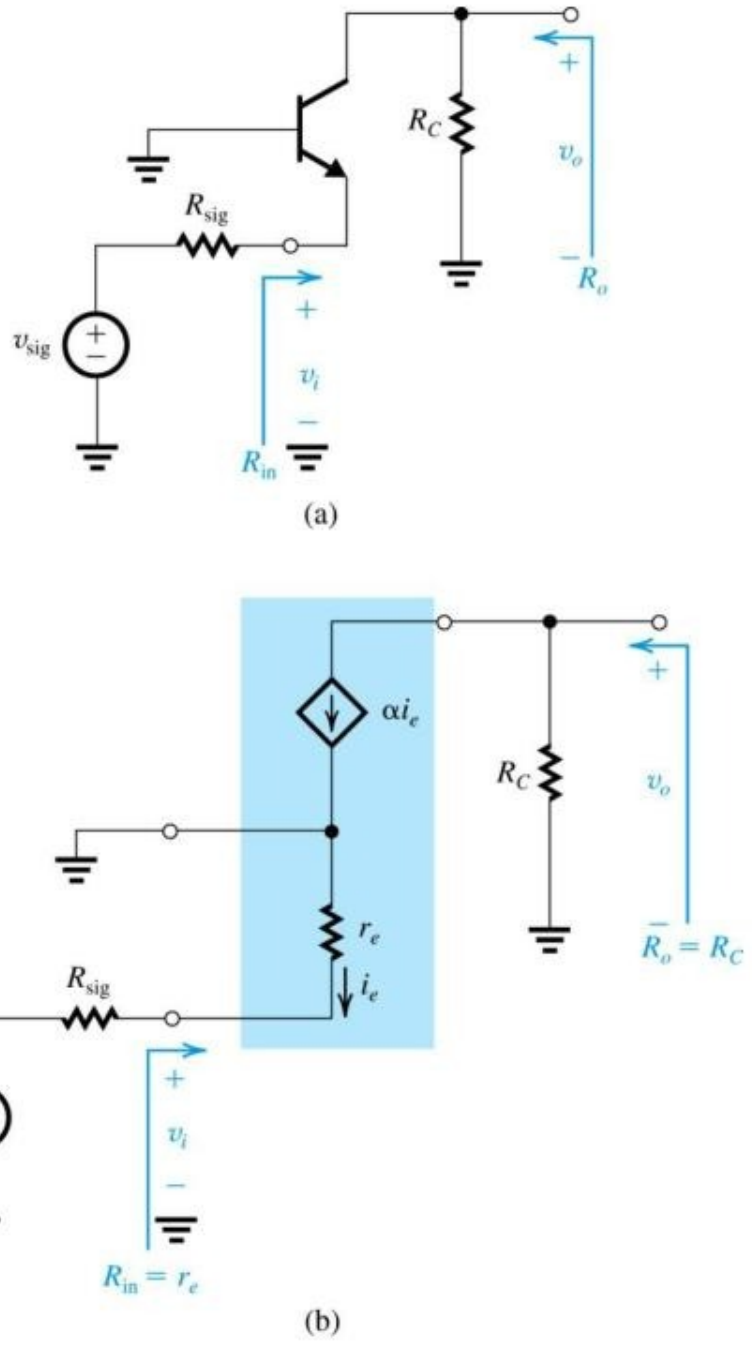


Figure 6.53: (**a**) CB amplifier with bias details omitted; (**b**) Amplifier equivalent circuit with the BJT represented by its T Model.

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6.6.7. **Summary and Comparisons**

- The CE configuration is one of the best suited for realizing the bulk of the gain required in an amplifier. Depending on the magnitude of the gain required, either a single stage o a cascade of two or three stages may be used.
- Including a resistor R_e in the emitter lead of the CE stage provides a number of performance improvements at the expense of gain reduction.
- The low input resistance of the CB amplifier makes it useful only in specific applications.
- The emitter follower finds application as a voltage buffer for connecting a high resistance source to a low-resistance load.

TABLE 6.5 Characteristics of BJT Amplifiers[a, b, c]

	R_{in}	A_{vo}	R_o	A_v	G_v
Common emitter (Fig. 6.50)	$(\beta+1)r_e$	$-g_mR_C$	R_C	$-g_m(R_C \parallel R_L)$ $-\alpha\dfrac{R_C \parallel R_L}{r_e}$	$-\beta\dfrac{R_C \parallel R_L}{R_{sig}+(\beta+1)r_e}$
Common emitter with R_e (Fig. 6.52)	$(\beta+1)(r_e+R_e)$	$-\dfrac{g_mR_C}{1+g_mR_e}$	R_C	$\dfrac{-g_m(R_C \parallel R_L)}{1+g_mR_e}$ $-\alpha\dfrac{R_C \parallel R_L}{r_e+R_e}$	$-\beta\dfrac{R_C \parallel R_L}{R_{sig}+(\beta+1)(r_e+R_e)}$
Common base (Fig. 6.53)	r_e	g_mR_C	R_C	$g_m(R_C \parallel R_L)$ $\alpha\dfrac{R_C \parallel R_L}{r_e}$	$\alpha\dfrac{R_C \parallel R_L}{R_{sig}+r_e}$
Emitter follower (Fig. 6.55)	$(\beta+1)(r_e+R_L)$	1	r_e	$\dfrac{R_L}{R_L+r_e}$	$\dfrac{R_L}{R_L+r_e+R_{sig}/(\beta+1)}$ $G_{vo}=1$ $R_{out}=r_e+\dfrac{R_{sig}}{\beta+1}$

[a] For the interpretation of R_{in}, A_{vo}, and R_o refer to Fig. 6.49.
[b] The BJT output resistance r_o has been neglected, which is permitted in the discrete-circuit amplifiers studied in this chapter. For integrated-circuit amplifiers (Chapter 7), r_o must always be taken into account.
[c] Setting $\beta=\infty$ ($\alpha=1$) and replacing r_e with $1/g_m$, R_C with R_D, and R_e with R_s results in the corresponding formulas for MOSFET amplifiers (Table 5.4).

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Summary

- Depending on the bias condition on its two junctions, the BJT can operate in one of three possible modes:
  - **cut-off** (both junctions reverse biased)
  - **active** (the EBJ forward-biased and CBJ reversed)
  - **saturation** (both junctions forward biased)
- For amplifier applications, the BJT is operated in the active mode. Switching applications make use of the cutoff and saturation modes.
- A BJT operating in the active mode provides a collector current $i_C = I_S \mathbf{exp}\{v_{BE}/V_T\}$. The base current $i_B = i_C/\beta$, and emitter current $i_E = i_C + i_B$.

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Summary

- To ensure operation in the active mode, the collector voltage of an *npn*-transistor must be kept higher than approximately $0.4V$ below the base voltage. For a *pnp*-transistor, the collector voltage must be lower than approximately $0.4V$ above the base voltage. Otherwise, the CBJ becomes forward-biased and the transistor will enter saturation.
- At a constant collector current, the magnitude of the base emitter voltage decreases by about $2mV$ for every $1^{O}C$ rise in temperature.
- The BJT will be at the edge of saturation when $|v_{CE}|$ is reduced to about $0.3V$.

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Summary

- In the active mode, i_C shows a slight dependence on v_{CE}. This phenomenon, known as the Early Effect, is modeled by ascribing a finite output resistance to the BJT: $r_o = |V_A|/I'_C$ where V_A is the Early Voltage and I'_C is the dc collector current without the Early Effect taken into account.
- The dc analysis of transistor circuits is generally simplified by assuming $|V_{BE}| = 0.7V$.
- To operate as a linear amplifier, the BJT is biased in the active region and the signal v_{be} is kept small ($v_{be} << V_T$).
- Bias design seeks to establish a dc collector current that is as independent of β as possible.

Summary

- For small signals, the BJT functions as a linear voltage-controlled current source with transconductance $g_m = I_C/V_T$. The input resistance between base and emitter, looking into the base, is $r_\pi = \beta/g_m$. The input resistance between bae and emitter, looking into the emitter is $r_e = 1/g_m$.
- Three basic BJT amplifier configurations are shown in Figure 6.48. A summary of their characteristic parameters is provided in Table 6.5.